Learning When to Warn, When to Ban: Constrained Reinforcement Learning for Multilingual Content Moderation

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Abstract

We frame multilingual Discord content moderation as a Constrained Markov Decision Process (CMDP). The agent learns a graduated action policy ALLOW, WARN, DELETE, TIMEOUT, BAN across 13 languages, with hard action masking enforcing procedural escalation (timeouts require prior infractions, bans require prior timeouts). The 403-dimensional observation combines multilingual sentence embeddings, per-language calibrated XLM-R toxicity scores, user infraction history with cool-down decay, server heat signals, and a language identity encoding. Fairness is enforced via an adaptive Lagrangian multiplier penalizing false positives and a Disparate Impact Wrapper that penalizes any language's ban rate exceeding $1.5\times$ the baseline. Training is a two-stage behavior-cloning to MaskablePPO pipeline. Trained on 595 synthetic episodes (8,016 steps), the agent achieves 91.7% accuracy on DELETE/TIMEOUT/BAN categories where static baselines score 0–4.5% with false-negative rate below 10^{-4}. Per-language parity holds for 12 of 14 categories; residual Hindi/Hinglish disparate impact is traced to upstream XLM-R bias.

1 Introduction

Online communities face an escalating challenge in managing toxic behavior across linguistically diverse user bases. Automated moderation systems traditionally rely on keyword filters or static toxicity thresholds, which produce coarse binary decisions (remove or permit) and fail to capture the graduated nature of community enforcement [1]. A single ban applied without prior warning violates procedural norms users expect of moderation systems; conversely, allowing repeated borderline behavior because no single message clears a fixed threshold enables sustained harassment campaigns [2].

Reinforcement learning offers a different framing: an agent that learns when to warn, delete, temporarily suspend, and permanently ban based on the full trajectory of a user's behavior and the toxicity of each message. Prior RL approaches have been largely English-centric [3] and have not addressed the fairness problem that arises when classifier scores are systematically biased across languages [4].

We take this alternative seriously and frame moderation as a Constrained Markov Decision Process. A single MaskablePPO policy operates over a five-level escalation ladder (ALLOW, WARN, DELETE, TIMEOUT, BAN) and is shared across 13 languages, removing the need for per-language models and making cross-language fairness comparisons meaningful. Two design decisions distinguish the approach. First, the procedural escalation rules—bans require prior timeouts, timeouts require prior infractions—are enforced as hard action masks rather than reward penalties, so the policy cannot even propose an illegal action. Second, fairness is treated as a training-time objective rather than a post-hoc audit: an adaptive Lagrangian multiplier suppresses false positives while a Disparate Impact Wrapper taxes any language whose ban rate runs above $1.5\times$ the baseline.

Submitted to the 1st Maynooth University Reinforcement Learning Conference (MURL 2026). Do not distribute.

The systematic per-language bias of the underlying XLM-R toxicity classifier is neutralized by an affine recalibration applied before scores reach the policy, and a language-aware regex layer overrides under-detected explicit threats so BAN remains available when the classifier is unreliable. Training proceeds in two stages—behavior cloning from an oracle escalation rule, followed by MaskablePPO fine-tuning—with an escalation-consistency reward shaping trajectory quality beyond per-step accuracy. On a held-out test split, the agent is the only system tested that reaches TIMEOUT and BAN correctly, and a residual fairness audit isolates a Hindi/Hinglish ban-rate gap and attributes it to upstream classifier bias rather than policy bias. Source code: `https://gitlab.cs.nuim.ie/2026/cs637/proj/fairmod`.

2 Graduated moderation as a constrained MDP

We model moderation as a CMDP $\mathcal{M} = (S, A, P, R, \mathbf{C}, \mathbf{d}, \gamma)$ with observation space $S \subseteq \mathbb{R}^{403}$, discrete action space $A = \{0, 1, 2, 3, 4\}$ corresponding to {ALLOW, WARN, DELETE, TIMEOUT, BAN}, transition kernel P, reward R, a vector $\mathbf{C}$ of K constraint cost functions with thresholds $\mathbf{d}$, and discount factor $\gamma = 0.95$. The objective is to maximize the return $V_R^\pi = \mathbb{E}_\pi[\sum_t \gamma^t R_t]$ subject to $V_{C_k}^\pi \leq d_k$ for $k = 1, \ldots, K$. The constraints encode fairness properties (false-positive rate; per-language ban-rate parity) that we want the trained policy to respect, not merely optimize against on average.

We solve the CMDP via Lagrangian relaxation [5, 14]. The primal–dual objective $\mathcal{L}(\pi, \boldsymbol{\lambda}) = V_R^\pi - \sum_k \lambda_k (V_{C_k}^\pi - d_k)$ reduces to a standard MDP with augmented per-step reward $\tilde{R}_t = R_t - \sum_k \lambda_k C_{k,t}$. The dual variable is updated by projected gradient ascent on the constraint violation,

$$\lambda_{k+1} = \max\Big(\epsilon,\ \lambda_k + \alpha_\lambda(\hat{V}_{C_k} - d_k)\Big)\,, \tag{1}$$

with $\hat{V}_{C_k}$ estimated over a 1,000-step sliding window, $\alpha_\lambda = 0.01$, and $\epsilon = 0.01$ preventing λ_k collapse.

Reward. The base reward R_t^{base} depends on the distance between the chosen action a_t and a ground-truth best action a_t^* derived from message toxicity and user history,

$$R_t^{\text{base}} = \begin{cases} +1.5 & a_t = a_t^* \\ -0.5 & |a_t - a_t^*| = 1 \\ -|a_t - a_t^*| & \text{otherwise.} \end{cases} \tag{2}$$

Context-sensitive penalties augment the base term: -3.0 for BAN without a prior timeout, -2.0 for TIMEOUT with $e_t < 2$, -2.0 for any punitive action on benign content ($s_t < 0.20$), and -3.0 for ALLOW on clearly toxic content ($s_t \geq 0.70$). The total is scaled and clipped to $[-1, 1]$, stabilizing the PPO value network [6].

3 Hard procedural enforcement via action masking

A central design decision is to enforce the procedural escalation ladder at the action-space level rather than through reward shaping alone. This is a stronger guarantee: a learned policy cannot, even by accident, propose an illegal action because masked actions are removed from the policy's distribution before sampling. We implement this with a binary mask $\mathbf{m}_t \in \{0, 1\}^5$ where $m_{t,a} = 0$ renders action a illegal at step t:

$$m_{t,4} = 0 \text{ if } h_t[\text{timeouts}] = 0 \tag{3}$$
$$m_{t,3} = 0 \text{ if } e_t < 2 \tag{4}$$
$$m_{t,a} = 0 \text{ if } s_t < 0.15,\ a \in \{1, 2, 3, 4\} \tag{5}$$
$$m_{t,0} = 0 \text{ if } s_t \geq 0.85 \tag{6}$$

Here s_t is the calibrated toxicity score and e_t the effective infraction count. BAN requires a prior timeout; TIMEOUT requires at least two prior infractions; punitive actions are forbidden on near-zero toxicity; and ALLOW is forbidden on clearly toxic content. Table 1 summarizes the action space.

Table 1: Action space and masking preconditions.

ID	Action	Precondition (must hold to be legal)
0	ALLOW	$s_t < 0.85$
1	WARN	$s_t \geq 0.15$
2	DELETE	$s_t \geq 0.15$
3	TIMEOUT	$s_t \geq 0.15,\ e_t \geq 2$
4	BAN	$s_t \geq 0.15,\ h_t[\text{timeouts}] \geq 1$

Table 2: Observation space components (403 total dimensions).

Symbol	Component	Dims	Description
$\mathbf{e}_t$	Message embedding	384	MiniLM-L12-v2 [13]
s_t	Toxicity score	1	Calibrated XLM-R score $\in [0, 1]$
$\mathbf{h}_t$	User history	3	Warns, timeouts, effective infractions
$\mathbf{q}_t$	Server context	2	Rolling toxicity rate, action rate
$\mathbf{l}_t$	Language identity	13	One-hot over 13 language codes

4 One multilingual policy across 13 languages

A single MaskablePPO policy is trained over an observation $\mathbf{o}_t = [\mathbf{e}_t \,\|\, s_t \,\|\, \mathbf{h}_t \,\|\, \mathbf{q}_t \,\|\, \mathbf{l}_t] \in \mathbb{R}^{403}$ that fuses signals from five sources (Table 2). Putting all 13 languages through a shared policy rather than per-language models makes fairness constraints meaningful: the Disparate Impact Wrapper (Section 6) compares ban rates *across* languages within the same policy, which would be incoherent if each language had its own model.

5 Per-language calibration of toxicity scores

Raw XLM-R toxicity scores are not comparable across languages: Arabic averages 0.139 below English-equivalent content; Italian, Ukrainian, and German are also under-scored, while Hinglish is over-scored by $+0.095$ on average relative to English. Feeding such scores directly into a shared policy bakes the classifier's per-language bias into every downstream decision. We address this with a per-language affine recalibration. For each language l, let s^l_{low} and s^l_{high} be the 75th percentiles of benign and toxic message scores in the training corpus; English anchors serve as the calibration target. The calibrated score is

$$\hat{s}(x, l) = \text{clip}(s_{\text{raw}}(x) \cdot \sigma_l + \beta_l,\ 0,\ 1), \tag{7}$$

with $\sigma_l = (s^{\text{en}}_{\text{high}} - s^{\text{en}}_{\text{low}})/(s^l_{\text{high}} - s^l_{\text{low}})$ and $\beta_l = s^{\text{en}}_{\text{low}} - s^l_{\text{low}} \cdot \sigma_l$. Calibration parameters are computed once and applied identically during training and inference, so no train/test distribution shift is introduced.

6 Dual fairness protection

We treat fairness as part of the optimization objective via two complementary mechanisms. The **LagrangianPenaltyWrapper** implements (1) with $\lambda_{\text{init}} = 0.3$, $\alpha_\lambda = 0.01$, and false-positive threshold $d = 0.05$ over a 1,000-step window; any punitive action with $s_t < 0.30$ counts as a false positive. The **DisparateImpactWrapper** monitors per-language BAN rates over a rolling 2,000-step window and applies a -2.0 penalty whenever a language l satisfies $\hat{\rho}_l > 1.5 \cdot \hat{\rho}_{\text{all}}$ and the current action is BAN. Together these turn fairness into a first-class training signal rather than a post-hoc audit. The wrappers are applied in a fixed order so that all penalty terms accumulate in raw reward space before the final scaling step bounds the total in $[-1, 1]$.

7 Two-stage training and escalation-consistency reward

Recent work on scaling RL for content moderation [15] reports that RL-only training without a behavioral prior is unstable: the policy fails to converge to a coherent reasoning schema. We therefore split training into two stages. In **Stage 1 (behavior cloning)**, an oracle rule $a_t^* = f(s_t, e_t, \tau_u)$ derived from the escalation ladder is rolled out across all training episodes and the MaskablePPO policy is trained via NLL minimization on the resulting $(\mathbf{o}_t, a_t^*, \mathbf{m}_t)$ triples for 30 epochs (batch size 256, learning rate 3×10^{-4}). In **Stage 2 (RL fine-tuning)**, the BC-initialized policy is loaded into MaskablePPO [7] from `stable-baselines3-contrib` [8] and fine-tuned for 10^6 timesteps. We use $n_{\text{steps}} = 4096$, batch size $= 256$, $n_{\text{epochs}} = 10$, $\gamma = 0.95$, learning rate 3×10^{-4}, entropy coefficient 0.01, max gradient norm 0.5. Four parallel environments yield an effective batch of $16{,}384$ transitions per update, well above the $\sim$1,024 inflection point identified by [15].

Beyond per-step accuracy, an *escalation-consistency* component shapes trajectory quality:

$$R_t^{\text{esc}} = \begin{cases} +0.3 & a_t > a_{t-1},\ s_t - s_{t-1} > 0.10 \\ +0.2 & a_t = 0,\ s_t < 0.20,\ c_t \geq 3 \\ -0.4 & a_t - a_{t-1} \geq 2,\ s_t - s_{t-1} < 0.20 \\ -0.3 & a_t > 0,\ c_t \geq 5,\ s_t < 0.30 \\ 0 & \text{otherwise.} \end{cases} \tag{8}$$

This rewards proportional escalation when toxicity rises, allows for de-escalation when a user has been clean, and penalizes both unjustified severity jumps and re-punishment of users who are visibly rehabilitating. The combined reward is $R_t = R_t^{\text{base}} + R_t^{\text{esc}}$.

8 Regex hard overrides for critical threats

Neural classifiers under-detect explicit threats with surprising regularity, so relying solely on $\hat{s}$ would leave a critical-harm gap. We add a language-aware regex layer covering five categories (stalking, violence, death wishes, swatting/doxxing, mass violence) across all 13 languages, using character-based matching for Japanese and Chinese and right-to-left-safe patterns for Arabic and Hebrew. When a pattern fires and $\hat{s} < 0.80$, the score is overridden to $\max(\hat{s}, 0.85)$, ensuring BAN remains available for explicit threats regardless of classifier confidence. Death-threat patterns originally assumed verb-before-object word order; adding SOV reverse-order variants for Russian, Ukrainian, Spanish, and Hindi raised threat coverage from 9/13 to 13/13 languages.

9 Synthetic data

Training episodes are generated in three stages: (i) a balanced corpus of $5{,}000$ toxic and $5{,}000$ benign examples per language is sourced from Jigsaw [11] and `textdetox/multilingual_toxicity_dataset` [10]; (ii) k-means clustering on MiniLM embeddings yields diverse per-language seeds; and (iii) DeepSeek V3.1 [12] is prompted via Baseten to synthesize multi-user Discord threads from those seeds. Thread types follow a fixed mix (30% toxic, 20% escalating, 15% mild, 10% subtle, 25% benign), with Hinglish skewed to 20%/30% benign/mild. The final dataset comprises 595 episodes (8,016 steps) across 14 language categories (Table 3) under an 80/20 train/test split.

10 Results

Reaching higher-severity actions. Table 4 compares the RL agent against two baselines on the held-out test split. The **keyword filter** triggers DELETE on hand-crafted matches; all else ALLOW. The **static threshold** maps toxicity ranges to fixed actions: $s < 0.15 \rightarrow$ ALLOW, $0.15 \leq s < 0.50 \rightarrow$ WARN, $s \geq 0.50 \rightarrow$ DELETE. Both baselines achieve 0% on TIMEOUT and BAN: they are fundamentally unable to reach the higher-severity actions because they have no notion of user history. The RL agent is the only system tested that produces correct TIMEOUT and BAN decisions, with per-class accuracies of 91.6% and 89.5% (Table 5). Its false-negative rate is 10^{-4}, near zero. The elevated false-positive rate (0.153) reflects conservative over-warning at the benign boundary; the

Table 3: Language distribution in evaluation corpus (8,016 steps).

Lang	Steps	%	Lang	Steps	%
hi	1014	12.6	uk	478	6.0
en	703	8.8	de	565	7.0
ar	687	8.6	zh	502	6.3
es	616	7.7	ru	484	6.0
ja	616	7.7	hin	400	5.0
fr	609	7.6	other	247	3.1
he	576	7.2	it	519	6.5

Table 4: Baseline performance comparison (8,016 steps, held-out test split).

System	Accuracy	FP rate	FN rate	TIMEOUT/BAN
Keyword filter	0.673	0.020	0.272	none
Static threshold	0.770	0.000	0.000	none
RL agent	**0.736**	0.153	$\mathbf{10^{-4}}$	**yes**

majority of false positives are WARN actions on benign content, a strictly lower-harm error than false DELETE or BAN.

Trajectory-level behavior. Three designed scenarios verify that the masking + escalation-consistency combination produces the intended trajectory-level behavior. **Escalation:** ten messages of monotonically increasing toxicity produce the full graduated ladder ALLOW^3 $\rightarrow$ WARN $\rightarrow$ DELETE $\rightarrow$ TIMEOUT^3 $\rightarrow$ BAN, respecting all masking preconditions. **Sustained troll:** a user alternating borderline-toxic and benign messages without ever triggering a single high-severity event is correctly escalated through recidivism (DELETE $\rightarrow$ ALLOW $\rightarrow$ DELETE $\rightarrow$ TIMEOUT^3), confirming the policy uses cumulative history rather than per-message severity.

Cross-lingual parity. Table 6 reports the XLM-R cross-lingual spread for six representative sentence categories. Three categories show spreads > 0.30: mild frustration (Arabic, Italian, Ukrainian, German under-detect), veiled threats (Ukrainian, Italian near zero while Japanese near one), and severe toxic content (Arabic 0.349 vs. near-unity elsewhere). These reflect fundamental limitations of the XLM-R pretrained weights rather than calibration failures: affine calibration cannot supply signal that the classifier does not produce.

Fairness audit and residual bias diagnosis. Table 7 reports per-language BAN rates (overall 0.066). Twelve of fourteen categories pass the $1.5\times$ parity threshold; Hindi ($1.92\times$) and Hinglish ($1.64\times$) exhibit residual disparate impact. Cross-referencing with Table 6 attributes this specifically to upstream classifier bias ($+0.095$ for Hindi vs. English) that compounds through the escalation ladder. Importantly, aggressive calibration offsets did not reduce the disparity, confirming the bias is structural to the classifier rather than the policy. This separates policy bias from model bias cleanly and points classifier fine-tuning [16] as the recommended remediation.

11 Conclusion

We presented a CMDP framing of multilingual content moderation in which the agent learns a graduated action policy across 13 languages, with action masking providing hard procedural enforcement and a Lagrangian/Disparate Impact stack making fairness a training-time objective. The two-stage BC$\rightarrow$RL pipeline plus an escalation-consistency reward yields trajectory-level reasoning, and language-aware regex overrides close the critical-harm gap left by classifier under-detection. The agent is the only system tested that produces correct higher-severity TIMEOUT and BAN actions, with 89–94% per-class accuracy and a near-zero false-negative rate. The fairness audit isolates residual Hindi/Hinglish disparate impact and attributes it to upstream classifier bias rather than policy bias, separating concerns for future work on classifier fine-tuning, disagreement-based reweighting [15], and multi-channel server graph extensions.

Table 5: RL agent per-class accuracy (held-out test split).

True action	Correct	Per-class accuracy
ALLOW	4305 / 5530	77.8%
WARN	185 / 953	19.4%
DELETE	507 / 539	94.1%
TIMEOUT	406 / 443	91.6%
BAN	493 / 551	89.5%
Overall	**5896 / 8016**	**73.6%**

Table 6: Cross-lingual toxicity score spread after calibration.

Category	Severity	Spread	Status
`benign_greeting`	benign	0.001	✓
`direct_insult`	moderate	0.067	✓
`hate_speech`	extreme	0.045	✓
`severe_toxic`	extreme	0.650	×
`mild_frustration`	mild	0.947	×
`threat_veiled`	threat	0.976	×

Table 7: Per-language BAN rate fairness audit (held-out test split).

Lang	Rate	Ratio	St.	Lang	Rate	Ratio	St.
`ar`	0.074	1.13	✓	`it`	0.004	0.06	✓
`de`	0.050	0.76	✓	`ja`	0.068	1.04	✓
`en`	0.048	0.74	✓	`ru`	0.072	1.10	✓
`es`	0.036	0.54	✓	`uk`	0.082	1.24	✓
`fr`	0.007	0.10	✓	`zh`	0.098	1.49	✓
`he`	0.078	1.19	✓	`hi`	0.126	1.92	×
`hin`	0.108	1.64	×				

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